Highlights

Discrete Learning Algorithms for Precipitation Estimation from Commercial Microwave Links

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- Rainfall reconstruction from CMLs is formulated as an explicit inverse optimization problem.
- A benchmark is built from EURADCLIM rain maps and synthetic link attenuations.
- The formulation preserves path geometry instead of collapsing links to point measurements.

Discrete Learning Algorithms for Precipitation Estimation from Commercial Microwave Links

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ABSTRACT

Commercial microwave links (CMLs) are part of the infrastructure of wireless networks. Their measured attenuations have been studied as an opportunistic source for monitoring spatiotemporal rainfall and other atmospheric phenomena. CML attenuation measurements can enhance the spatiotemporal accuracy and resolution of existing weather monitoring instruments. In addition, they serve as stand-alone weather monitoring devices in places where dedicated weather monitoring devices are scarce or do not exist.

Current techniques for 2D rainfall map reconstruction usually reduce CML measurements to virtual rain gauges, namely point measurements, and rely on interpolation techniques such as inverse distance weighting or Kriging. While effective in many scenarios, these methods are suboptimal because they do not address the modeling error introduced when a path-integrated attenuation measurement is replaced by a single-point rain-intensity value.

In this study, we revisit the rainfall map reconstruction problem from CML signal attenuation measurements with a principled optimization approach. We formulate the problem of partial-to-complete field reconstruction as a physics-informed optimization problem. The reconstructed rainfall field is quantized and represented by pixel rainfall variables whose values are constrained to agree with the observed CML signal attenuations. The resulting solution minimizes a weighted sum of attenuation errors along the links, spatial differences between neighboring pixels, and the total rainfall over the map.

To evaluate our approach, we create a benchmark of 200 rainfall maps and CML configurations, comprising 100 winter and 100 summer patches. Rainfall maps are extracted from EURADCLIM by identifying rain events across diverse terrain types and precipitation patterns. The selected patches have side lengths between 50 and 94 km. CMLs are overlaid using a publicly available Netherlands dataset, and attenuations are computed using the ITU-R P.838 model.

We then solve the inverse problem to reconstruct rainfall maps from attenuations and compare the results with inverse distance weighting. Overall, this study reframes rainfall reconstruction from opportunistic sensing networks as a well-posed inverse problem with an explicit objective function. The framework also provides interpretability benefits compared to purely data-driven approaches.

1. Introduction

Accurate rainfall estimation is a major tool used for accurate hydrology studies, weather prediction, flood forecasting, and urban drainage management. Reliable information on the spatial and temporal structure of precipitation is particularly important in urban and densely populated environments, where short-duration convective events may produce strong local gradients and rapid hydrological response. Conventional weather monitoring tools, primarily rain gauges and weather radars, provide valuable information but also have well-known limitations. Rain gauges offer direct point measurements, and are considered very accurate when properly installed, but their spatial density is often insufficient for resolving localized rainfall variability. Weather radars provide wide-area coverage, yet their estimates are not at ground level, and are prone to many uncertainties such as scattering, blockages, unaccounted vertical variability of rainfall, and limited near-ground representativeness.

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Commercial microwave links (CMLs) offer a complementary source of rainfall information. These links are widely deployed as part of cellular telecommunication backhaul networks and continuously transmit electromagnetic signals between fixed antennas in a line-of-sight installments, nowadays at frequencies of 10 GHz to 90 GHz. During rainfall, raindrops affect the microwave radiation, causing measurable attenuation of the received signal. Since the frequencies range of operational links are sensitive to rain-induced attenuation ITU-R (2005); Gunn and East (1954); Olsen et al. (1978), the same infrastructure for communication was shown to also serve as an opportunistic environmental sensing network Overeem et al. (2013); Chwala and Kunstmann (2019); Fencl et al. (2015). As the CMLs are already deployed at large scale, often form dense networks in urban and suburban areas (especially in recent times, where dedicated CMLs networks for smart-city management are being deployed Ostrometzky and Messer (2024); Janco et al. (2022), and are gaining attention as part of the integrated sensing and communication (ISAC) paradigm Han et al. (2023)), they can provide near-ground measurements integrated along their paths, which span between tens of meters to several kilometers. Note that CMLs can be used in combination with satellite microwave links (SML) to further enhance their capabilities Nebuloni et al. (2025).

The physical basis for CML rainfall sensing is based on the attenuation of the signal that is caused by precipitation Gunn and East (1954); Olsen et al. (1978), and is described by the ITU-R power-law relation AK: We should cite the standard here, bu between rain rate and specific attenuation. Given the frequency and polarization of a link, the rain-induced attenuation accumulated along the link can be modeled as the path integral of the specific attenuation over the rainfall field. Thus, unlike a rain gauge, which measures rainfall at a point, a CML provides a path-integrated measurement. This distinction is fundamental. A single link does not directly observe the rainfall value at one location; rather, it constrains the average attenuation, and hence the rainfall intensity, along its propagation path. Consequently, rainfall reconstruction from CMLs is an inverse problem: the objective is to estimate a spatial rainfall field from a collection of line-integral-like measurements.

Many practical CML-based reconstruction methods simplify this inverse problem by first converting each link measurement into a single representative rain-rate value, often interpreted as a virtual rain gauge located at the link midpoint or at an assumed other point along the link path. Standard spatial interpolation techniques can then be applied to these virtual point observations. This approach is simple, computationally efficient, and often useful in practice. However, it also discards part of the geometric information contained in the link measurement. In particular, the fact that the attenuation is accumulated along an extended path is replaced by a point approximation. This simplification may introduce modeling errors, especially in heterogeneous rainfall fields or in regions where the geometry and density of the link network strongly influence the information available for reconstruction Gat and Messer (2019); Leijnse et al. (2008).

A more faithful treatment of CML data should account for the coupling between the unknown rainfall field and the measured attenuation along each link path. This coupling depends on the link geometry, length, frequency, polarization, and the spatial distribution of rain along the path. Reconstruction methods that explicitly incorporate this structure have the potential to use the available information more effectively, but they also lead to more challenging estimation problems. In particular, because the attenuation–rain-rate relation is nonlinear, the corresponding inverse problem may be formulated as a constrained nonconvex optimization problem over the rainfall field.

Evaluating such reconstruction methods is difficult in real operational settings because the true rainfall field is generally unknown. Real CML measurements also include additional effects that complicate the analysis, such as wet-antenna attenuation, baseline drift, quantization, hardware-dependent sampling, and measurement noise. While these effects are essential in operational systems, they make it difficult to isolate the intrinsic performance of the reconstruction algorithm itself. A controlled benchmark, in which the ground-truth rainfall field is known and the corresponding CML attenuations are generated through a forward physical model, provides a useful way to study this core inverse problem.

In this paper, we introduce such a benchmark framework. Starting from hourly EURADCLIM rainfall maps, we construct synthetic but physically grounded rain-event patches and generate corresponding noise-free CML attenuation measurements using the ITU-R forward model. This creates a controlled evaluation setting in which different reconstruction algorithms can be compared directly against the known rainfall field. By removing confounding measurement artifacts while preserving realistic spatial rainfall patterns, the proposed benchmark isolates the reconstruction problem and enables systematic analysis of the role of link geometry, interpolation assumptions, and optimization-based estimation.

We use this framework to compare standard inverse-distance weighting (IDW), a link-aware extension tailored to CML measurements, and a nonconvex optimization formulation that directly couples the estimated rainfall field to the

Table 1
ITU-R P.838-3 coefficients for horizontal polarization and resulting specific attenuation γ_R for selected rain rates ($\mathrm{mm\,h^{-1}}$).

f (GHz)	κ_H	α_H	$\gamma_R(R=1)$ ($\mathrm{dB\,km^{-1}}$)	$\gamma_R(R=2)$ ($\mathrm{dB\,km^{-1}}$)	$\gamma_R(R=4)$ ($\mathrm{dB\,km^{-1}}$)	$\gamma_R(R=8)$ ($\mathrm{dB\,km^{-1}}$)
10	0.012	1.257	0.012	0.029	0.070	0.166
15	0.045	1.123	0.045	0.098	0.213	0.463
20	0.092	1.057	0.092	0.191	0.397	0.825
25	0.157	0.999	0.157	0.314	0.628	1.254
30	0.240	0.949	0.240	0.464	0.895	1.727
40	0.443	0.867	0.443	0.808	1.475	2.690

measured attenuations. AK: We should cite something for the other two methods In addition, we analyze reconstruction quality as a function of distance from the available links, thereby quantifying how the spatial arrangement of the CML network affects estimation accuracy.

1.1. Contributions

1. Introduction of a methodology for generating rain-event benchmarks for evaluation of rainfall estimation based on noise-free CML attenuation.
2. Creation of a 200-patch benchmark comprising 100 winter and 100 summer patches.
3. Introduction of a new algorithm ILDW - version of IDW tailored for CML's.
4. Introduction of rain-rate estimation as a nonconvex constrained optimization problem. Improve coupling between the estimation and measured attenuation.
5. Systematic evaluation of IDW, ILDW, and nonconvex optimization algorithms.
6. Evidence of quality of estimates as a function of distance from links.

2. Background and Related Work

2.1. ITU-R Forward Model: Rain-Induced Attenuation

During a rain event, the signal along a commercial microwave link experiences attenuation because raindrops absorb and scatter electromagnetic energy. We use Recommendation ITU-R P.838-3 for converting rain-rate into specific attenuation (ITU-R, 2005). The ITU-R model relates the rain rate R (in $\mathrm{mm\,h^{-1}}$) to the *specific attenuation* γ_R (in $\mathrm{dB\,km^{-1}}$) through the power law

$$\gamma_R = \kappa R^{\alpha}, \tag{1}$$

where the coefficients κ and α depend on the link frequency and polarization (ITU-R, 2005). See Table 1 for the coefficient values and resulting attenuation for horizontal polarization and rain rates of 1–8 $\mathrm{mm\,h^{-1}}$.

The total rain-induced attenuation is the integral of γ_R along the link. Thus, longer links accumulate more attenuation, and links operating at different frequencies or polarizations respond differently to the same rainfall field. For a horizontal terrestrial link of length L with uniform rain rate R along the path, the total rain-induced attenuation is therefore $A = \gamma_R \cdot L$. If the rain-rate field is discretized into pixels, the total rain-induced attenuation is the sum of the attenuations over the segments obtained by intersecting the link with the pixels it traverses.

3. Benchmark Construction

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In this section, we describe how we constructed our benchmark of rain-event patches combined with CML rain-attenuation. The resulting benchmark instances combine realistically derived rainfall maps with real-world link configurations, from which noise-free rain-induced attenuation measurements are generated for every link.

3.1. Data Source

We extract rain events from the European climatological gauge-adjusted radar precipitation dataset EURADCLIM (Overeem et al., 2023). EURADCLIM is derived from OPERA European radar composites and provides hourly precipitation fields at 2 km spatial resolution over a large fraction of Europe. The dataset are distributed in HDF5 format according to the OPERA Data Information Model (ODIM) convention (EUMETNET OPERA, 2026). File-level access and documentation are provided through the KNMI data platform (KNMI, 2023).

Patch size (km)	# links	1-multiplicity	2-multiplicity
50×50	939	99	420
62×60	1254	128	563
72×70	1507	157	675

Table 2
Representative link multiplicities for three patch dimensions. The multiplicity of a link is defined as the number of frequencies assigned to it. The second column reports the total number of link–frequency pairs. In the data set, link multiplicities were either 1 or 2, with approximately 10% of links having multiplicity 1.

3.2. Automatic Extraction of Rain Events

Each hourly EURADCLIM map is processed independently. We first average the rainfall field over a 4×4 window of native 2 km pixels, then retain the pixels whose smoothed rain rate is at least 3 $\mathrm{mm\,h^{-1}}$. The retained pixels are partitioned into connected components under 8-neighborhood connectivity. The event-selection criterion requires both side lengths of a component's axis-aligned bounding box to fall between 50 km and 95 km; in the resulting 200-patch benchmark, the observed side lengths range from 50 to 94 km. The accepted bounding box defines the extent of the benchmark patch, so patch dimensions follow the rain events themselves. Fixing a common patch size would instead require cropping the larger events or padding the smaller ones with dry area.

3.3. Refinement and Smoothing

We refine and smooth each rain patch derived from EURADCLIM rainfall maps by subdividing each pixel into 16×16 subpixels (each 125×125 $\mathrm{m^2}$), followed by Gaussian smoothing with $\sigma = 1$ in subpixel units. The resulting rain patches are treated as the ground-truth rainfall field.

3.4. CML Dataset

CML metadata (including link locations, lengths, and frequencies) are obtained from the publicly available 4TU dataset (Overeem et al., 2016) (we assumed horizontal polarization for all links). For each rainfall patch, the set of links is cropped to the patch extent and overlaid on the rainfall field. The ITU-R rain-attenuation model is then applied along each link path to compute the corresponding attenuation values.

4. Benchmark Statistics

In this section, we summarize key properties of the benchmark patches and their associated link geometry. We describe the geographic distribution and size range of the selected rain events, illustrate representative patch instances, summarize link statistics and topology, and report how rainy and dry pixels are distributed with respect to distance from the third-closest link.

Patches.

Time. The rain patches were extracted from hourly EURADCLIM maps between 18 and 31 January 2023 (winter) and from June 2023 (summer).

Geographic distribution. See Figure 1 for a depiction of the geographic locations of all 200 rain patches.

Patch sizes. A scatter diagram of patch side lengths is depicted in Figure 2. The selected benchmark patches have side lengths ranging from 50 to 94 km.

Patch statistics.

Sample patches. Sample rain-events from the benchmark are depicted in Figures 7 and 9 together with their geographic location.

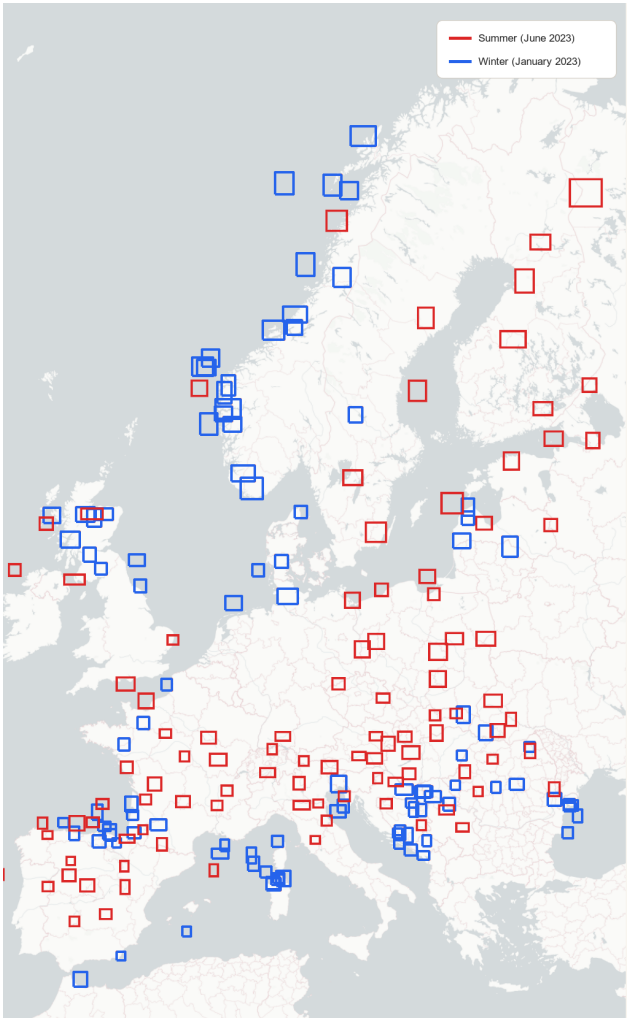


Figure 1: Geographic locations of the 100 summer patches (red) and 100 winter patches (blue).

Links. The largest patch overall by area is a summer patch from 19 June 2023 at 22:00, measuring 88×94 km. Its link geometry appears in Figure 3. The pixels in this figure are binned according to their distance to the third-closest link. For a pixel p, let $d_3(p)$ denote the distance from the pixel center to the third-closest link. Equivalently, $d_3(p)$ is the smallest radius such that a disk centered at p intersects at least three links.

Table 4 summarizes link lengths, frequencies and endpoint degrees in this patch (endpoint degree is the number of link–frequency observations incident to the endpoint). See Table 2 for the number of links and their multiplicities for typical patch sizes (multiplicity 2 means a link operating at two frequencies).

Rainy/Dry Pixel Distance Stratification. We considered all 200 benchmark patches and classified the pixels in each patch as rainy or dry using a rainfall threshold of $0.6\,\mathrm{mm\,h^{-1}}$. Each of these two pixel sets was then partitioned into bins according to the pixel-level distance $d_3(p)$. The mean and standard deviation of the rainy-pixel and dry-pixel counts in each distance bin, computed across all patches, are reported in Table 5.

5. Inverse Solvers: From Link Attenuations to Rainfall Maps

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In this section, we present three solvers for the inverse problem: IDW, ILDW, and a nonconvex optimization model.

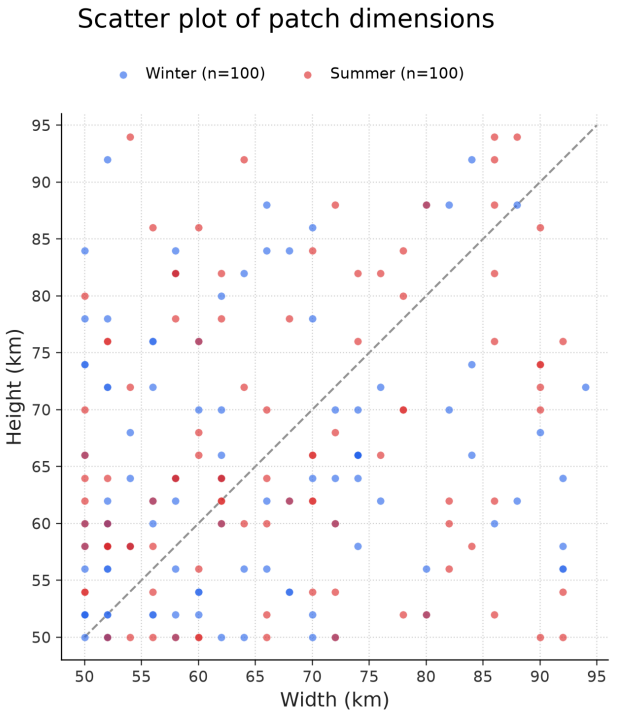


Figure 2: Patch width versus patch height for all 200 benchmark patches: winter patches are blue and summer patches are red. The mean patch area is 280,649 pixels (SD 83,607; pixel side length 125 m).

Descriptor	Mean	SD	Min	Median	Max
Wet fraction	0.803	0.157	0.329	0.816	1.000
Mean wet rain rate (mm/h)	4.49	1.61	2.39	3.99	9.97
95th percentile wet rain rate (mm/h)	12.19	7.07	3.48	9.80	34.23
Coefficient of variation, wet pixels	0.783	0.317	0.180	0.745	1.720
Area fraction above 5 mm/h	0.190	0.104	0.000	0.190	0.470
Relative gradient (km^{-1})	0.277	0.138	0.053	0.274	0.768

Table 3
Rain-pattern descriptors across all 200 benchmark patches, summarized from one value per patch. Wet pixels have rain rate at least 0.6 mm/h. For each patch, the wet-pixel coefficient of variation (CV) is the standard deviation of wet-pixel rain rates divided by their mean.

Relative gradient is defined as follows. At each pixel, the local gradient magnitude is $\sqrt{g_x^2 + g_y^2}$. Let $h = 0.125$ km be the spacing between adjacent pixel centers. At an interior pixel p, $g_x(p) = \frac{1}{2}\big[(R(p) - R(p_{\leftarrow}))/h + (R(p_{\rightarrow}) - R(p))/h\big]$, averaging the slope from the left neighbor to p and the slope from p to its right neighbor. Similarly, $g_y(p) = \frac{1}{2}\big[(R(p) - R(p_{\uparrow}))/h + (R(p_{\downarrow}) - R(p))/h\big]$, using the pixel above and the pixel below. At an edge where a neighbor is missing, the available one-sided slope is used. Thus both components have units of $\mathrm{mm\,h^{-1}\,km^{-1}}$. Dividing their patch-average magnitude by the patch mean rain rate cancels the rain-rate unit, so the relative gradient reported in the table has units of km^{-1}.

5.1. Problem Formulation

The inputs for the inverse problem consists of a set of links $\mathcal{L}$ in a patch P (set of pixels). Every link ℓ has an attenuation A_ℓ and the following attributes: frequency, polarization, endpoints and length $|\ell|$.

The output is an estimated rain-rate field $\hat{R} : P \to \mathbb{R}_{\geq 0}$.

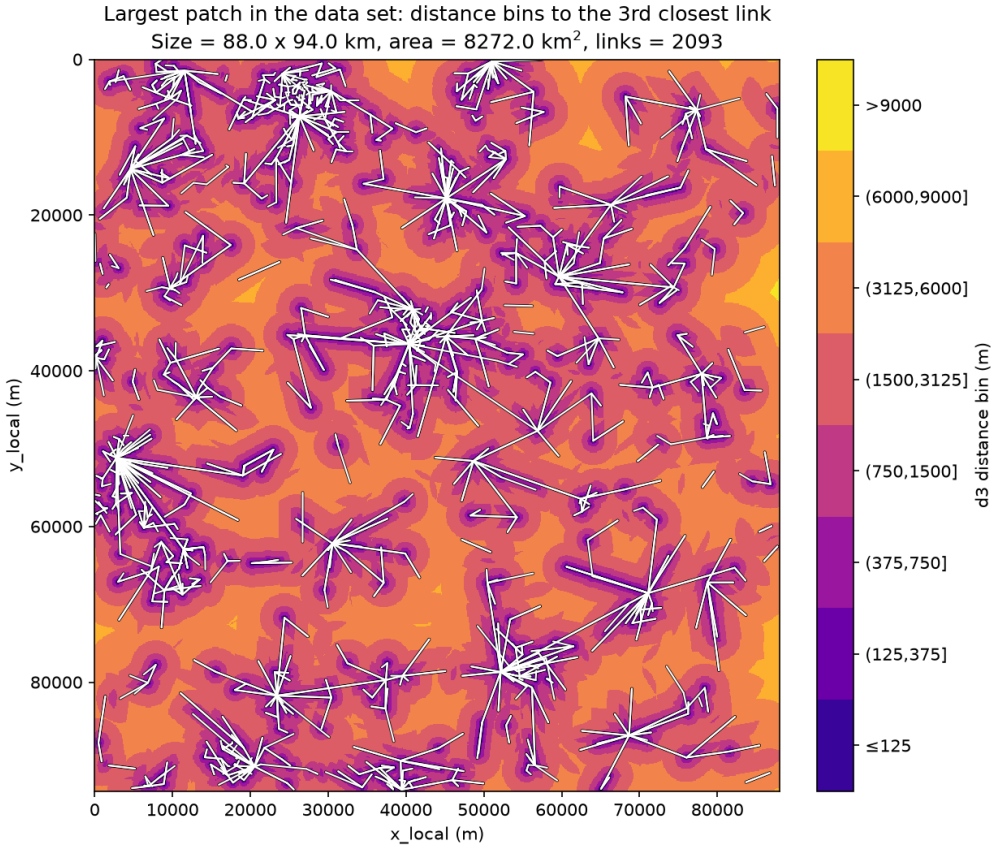


Figure 3: d_3-bin map for the largest patch overall by area: the summer patch from 19 June 2023 at 22:00 (88×94 km). Each pixel is colored according to its distance bin, and the complete link geometry is overlaid in white.

Length bin (km)	Count	%
$[0, 1)$	366	17.5
$[1, 2)$	571	27.3
$[2, 4)$	608	29.0
$[4, 8)$	423	20.2
$[8, 16)$	118	5.6
$[16, 24]$	7	0.3

Frequency bin (GHz)	Count	%
$[10, 15)$	20	1.0
$[15, 20)$	74	3.5
$[20, 25)$	87	4.2
$[25, 30)$	284	13.6
$[30, 35)$	106	5.1
$[35, 40)$	1433	68.5
$[40, 60]$	89	4.3

Endpoint-degree bin	Count	%
1	188	15.1
2	617	49.7
3, 4	268	21.6
5, 6	78	6.3
7..10	45	3.6
11..20	28	2.3
≥ 21	18	1.4

Table 4
Link statistics for the largest patch overall by area: the summer patch from 19 June 2023 at 22:00 (88×94 km; 2,093 link–frequency observations). The first panel counts observations by path-length interval, the second by carrier-frequency interval, and the third counts 1,242 endpoints by degree.

5.2. Rain Equivalent for Link Attenuation

The *rain-rate equivalent* R_ℓ of link ℓ is the uniform rain-rate along the link that would cause the link total attenuation A_ℓ. This rain-rate is derived from the ITU-R model by:

$$R_\ell \triangleq \left(\frac{A_\ell / |\ell|}{\kappa_\ell} \right)^{1/\alpha_\ell} ,$$

where $(\kappa_\ell, \alpha_\ell)$ the ITU-R P.838-3 coefficients corresponding to the link's frequency and polarization.

	Rainy pixels			Non-rainy pixels		
d_3 bin (m)	Mean count	Mean %	SD	Mean count	Mean %	SD
≤125	3905.3	1.80	1067.8	1053.3	1.67	967.8
(125,375]	15353.2	7.05	4225.3	4183.0	6.60	3763.6
(375,750]	24845.0	11.29	7280.0	6401.2	10.26	5763.7
(750,1500]	48587.4	21.88	15505.5	12184.0	18.92	11231.5
(1500,3125]	83847.9	37.38	29719.7	20272.2	31.78	19209.7
(3125,6000]	42951.5	18.98	16795.7	11518.4	19.75	11019.5
(6000,9000]	3314.4	1.46	2567.1	1631.6	3.61	2045.8
>9000	385.8	0.17	787.5	214.9	0.41	573.9
Total	223268.6	100	74239.5	57324.8	100	52090.6

Table 5
Ground-truth rainy and non-rainy pixel counts by d_3 bin. Counts and within-class percentages are first computed per patch and then averaged across all 200 patches.

5.3. IDW

Inverse-distance weighting (IDW) is one of the standard interpolation strategies used in commercial-microwave-link rainfall reconstruction (Zimmerman et al., 1999; Guo-Dong et al., 2003; Chen and Liu, 2012; Chwala and Kunstmann, 2019; Eshel et al., 2020, 2021). The input consists of a set of links, where each link ℓ carries a total attenuation, frequency, and polarization.

IDW for CML's replaces every link ℓ with a virtual rain-gauge g_ℓ positioned at the midpoint between the endpoints of ℓ. The rain-rate associated with g_ℓ is simply the rain-rate equivalent R_ℓ. Inverse-distance weighting is applied with the virtual gauges positions and rain-rates as inputs.

We consider a setting in which the weight of a gauge decreases quadratically (i.e., decay exponent equals 2) and the maximum radius of influence $r_{\max}$ is set to 15 km. Distances to pixels are measured to the center of the pixel. If one or more link midpoints lie inside pixel p, then the estimated rainfall in pixel p is set to the mean of their rainfall estimates. The pseudocode for IDW appears in Algorithm 1.

Algorithm 1: Discretized IDW. Each link is replaced with a virtual gauge located at the midpoint of the link. The rain-rate at the gauge is set to the rain-rate equivalent of its link. IDW is then applied with respect to these virtual gauges. We set the exponent of the decay to 2.

Data: link set $\mathcal{L}$, a patch P, and a maximum influence distance $r_{\max}$
Result: rainfall field $\hat{R} : P \to \mathbb{R}_{\geq 0}$

foreach *link* $\ell \in \mathcal{L}$ **do** // compute rain-rate equivalent per link
  $(\ell.\kappa, \ell.\alpha) \leftarrow \mathrm{ITU838}(\ell.\mathrm{freq}, \ell.\mathrm{polarization})$
  $R_\ell \leftarrow \left(\frac{A_\ell / |\ell|}{\ell.\kappa} \right)^{1/\ell.\alpha}$
foreach *pixel* $p \in P$ **do** // compute rain $\hat{R}(p)$
  $M_p \leftarrow \{\ell \in \mathcal{L} : \ell.\mathrm{midpoint} \in p\}$
  if $M_p \neq \emptyset$ **then**
    $\hat{R}(p) \leftarrow \frac{1}{|M_p|} \sum_{\ell \in M_p} R_\ell$ // pixel contains a link endpoint
  else
    $N_p \leftarrow \{\ell \in \mathcal{L} : \|\ell.\mathrm{midpoint} - p.\mathrm{center}\|_2 \leq r_{\max}\}$
    if $N_p = \emptyset$ **then**
      $\hat{R}(p) \leftarrow 0$ // pixel far from links
    else
      $\hat{R}(p) \leftarrow \sum_{\ell \in N_p} \frac{\|\ell.\mathrm{midpoint} - p.\mathrm{center}\|_2^{-2}}{\sum_{\ell' \in N_p} \|\ell'.\mathrm{midpoint} - p.\mathrm{center}\|_2^{-2}} \cdot R_\ell$

5.3.1. ILDW

We propose a variation of IDW that alleviates the mismatch between measured link attenuation and attenuation computed from the IDW-estimated rainfall field. We refer to this variation as inverse link-distance weighting (ILDW). ILDW assigns a weight to every link that is inversely proportional to the distance between the pixel and the link, where

$$\mathrm{dist}(p, \ell) \triangleq \min_{q \in \ell} \|p.\mathrm{center} - q\| \ .$$

As in IDW, we set the decay exponent to 2 and the maximum radius of influence $r_{\max}$ is set to 15 km. If one or more links cross the pixel, the estimated rain-rate is taken to be the mean of the corresponding link rain-rate equivalents. Otherwise, the estimate is a weighted average over links within distance $r_{\max}$ (set to 15km). The pseudocode for ILDW is shown in Algorithm 2, with lines that differ from IDW highlighted in red.

Algorithm 2: Discretized ILDW. Rainfall is averaged by inverse squared distance to a link. Lines that are different from their corresponding lines in IDW are colored red.

Data: link set $\mathcal{L}$, a patch P, and a maximum influence distance $r_{\max}$
Result: rainfall field $\hat{R} : P \to \mathbb{R}_{\geq 0}$

foreach *link* $\ell \in \mathcal{L}$ **do** // compute rain-rate equivalent per link
  $(\ell.\kappa, \ell.\alpha) \leftarrow \mathrm{ITU838}(\ell.\mathrm{freq}, \ell.\mathrm{polarization})$
  $R_\ell \leftarrow \left(\frac{A_\ell / |\ell|}{\ell.\kappa} \right)^{1/\ell.\alpha}$
foreach *pixel* $p \in P$ **do** // compute rain $\hat{R}(p)$
  $X_p \leftarrow \{\ell \in \mathcal{L} : \ell \cap p \neq \emptyset\}$
  if $X_p \neq \emptyset$ **then**
    $\hat{R}(p) \leftarrow \frac{1}{|X_p|} \sum_{\ell \in X_p} R_\ell$ // at least one link traverses the pixel
  else
    $N_p \leftarrow \{\ell \in \mathcal{L} : \mathrm{dist}(p, \ell) \leq r_{\max}\}$
    if $N_p = \emptyset$ **then**
      $\hat{R}(p) \leftarrow 0$ // pixel far from links
    else
      $\hat{R}(p) \leftarrow \sum_{\ell \in N_p} \frac{\mathrm{dist}^{-2}(p, \ell)}{\sum_{\ell' \in N_p} \mathrm{dist}^{-2}(p, \ell')} \cdot R_\ell$

5.3.2. Nonconvex Formulation

We formulate the inverse problem as a constrained optimization problem. The decision variables are the estimated rain-rates $\hat{R}(p)$ for all pixels $p \in P$. The constraints are simply nonnegativity constraints that reflect the physical requirement that rainfall is nonnegative. The objective is to minimize $J(\hat{R})$, where the cost function J consists of the 4 weighted terms:

$$J(\hat{R}) = w_{\text{atten}} \cdot J_{\text{atten}}(\hat{R}) + w_{1d} \cdot J_{1d}(\hat{R}) + w_{2d} \cdot J_{2d}(\hat{R}) + w_{\text{total}} \cdot J_{\text{total}}(\hat{R}) \, .$$

The terms in $J(\hat{R})$ are as follows:

1. The attenuation term J_{atten} quantifies disagreement between observed and estimated link attenuations. For a rain-rate field estimate $\hat{R} : P \to \mathbb{R}_{\geq 0}$ and a link ℓ, let $A_\ell(\hat{R})$ denote the attenuation induced on ℓ by $\hat{R}$. This attenuation is computed using the ITU-R forward model with the link's frequency and polarization. The attenuation term J_{atten} is defined by

$$J_{\text{atten}}(\hat{R}) \triangleq \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \frac{\left(A_\ell(\hat{R}) - A_\ell\right)^2}{|\ell|} \, .$$

2. The first-difference term J_{1d} enforces spatial smoothness, and is defined by

$$J_{1d}(\hat{R}) \triangleq \frac{1}{|P|} \sum_{(u,v) \in \mathcal{E}(P)} (\hat{R}(u) - \hat{R}(v))^2 \, ,$$

   where $\mathcal{E}(P)$ is the set of horizontally and vertically adjacent pixel pairs in patch P.
3. The second-difference term J_{2d} penalizes curvature, and is defined by

$$J_{2d}(\hat{R}) = \frac{1}{|P|} \sum_{(p,q,t) \in \mathcal{T}(P)} \left((\hat{R}(q) - \hat{R}(p)) - (\hat{R}(t) - \hat{R}(q))\right)^2,$$

   where $\mathcal{T}(P)$ is the set of horizontal and vertical triples of collinear pixels in patch P.
4. The shrinkage term J_{total} biases the solution toward smaller overall rainfall values, and is defined by

$$J_{\text{total}}(\hat{R}) \triangleq \frac{1}{|P|} \sum_{p \in P} \hat{R}(p)^2.$$

Implementation Details:

1. The weight of each term is set to the reciprocal of its value evaluated at the ILDW solution. We further divide w_{2d} by 2 and w_{total} by $|P|$. This normalizes the objective instance-by-instance.
2. We use the L-BFGS-B algorithm (Byrd et al., 1995; Zhu et al., 1997), a limited-memory quasi-Newton method for smooth constrained optimization, as implemented in SciPy (Virtanen et al., 2020).

Nonconvex Cost Function: The ITU-R forward model (Equation (1)) attaches an attenuation exponent α to every link based on its frequency and polarization. When $\alpha = 1$, the predicted attenuation is linear in the rain-rate variables, so J_{atten} is a sum of squares of linear residuals and is convex. For $\alpha \neq 1$, the power-law attenuation makes the residual nonlinear, and the attenuation term is generally nonconvex. The nonconvexity emphasizes the importance of the initial solution used by the solver because local minima need not be global minima. We illustrate this nonconvexity with a concrete 3×3 single-link example in Appendix A.1.

Summary of the Nonconvex Inverse Solver. The input to the inverse solver is a patch P and a set $\mathcal{L}$ of links. We treat each link $\ell \in \mathcal{L}$ as a record containing its geometry and physical attributes that include its frequency f_ℓ, polarization, length $|\ell|$, and measured attenuation A_ℓ. The L-BFGS-B solver also takes an initial rain-rate field $\hat{R}_{\text{init}}$ as input. We write

$$\hat{R} \leftarrow \text{LBFGSB}(\mathcal{L}, P, \hat{R}_{\text{init}})$$

to denote one invocation of the solver and the resulting rain-rate field estimate.

Coping with nonconvexity. We use three strategies to deal with the nonconvexity of J_{atten}:

1. Use ILDW as the initial solution. We refer to this solver as SOLVER(ILDW).
2. We initialize the nonconvex solver with the ground-truth rainfall field. This configuration, denoted SOLVER(GT), serves as a best-case reference for the nonconvex formulation.
3. Obtain a convex surrogate objective by replacing each link with a virtual link that keeps the same geometry but changes its frequency to f^0, chosen so that the attenuation exponent equals 1 (i.e., $f^0 = 25\,\text{GHz}$ for horizontal polarization). We refer to this solver as CONVEX SOLVER.
4. Apply homotopy continuation to smoothly transition from the convex surrogate to the original nonconvex formulation. We refer to this solver as HOMOTOPY SOLVER.

We elaborate on the last two strategies below.

5.4. CONVEX SOLVER: Convex Surrogate Inverse Solver

We construct a virtual link set $\mathcal{L}^0$ from the original link set $\mathcal{L}$. For each link $\ell \in \mathcal{L}$, the corresponding virtual link $\ell^0 \in \mathcal{L}^0$ keeps the same geometry and length, but uses the frequency f^0 for which the ITU-R attenuation exponent equals 1.[1] The attenuation of the virtual link ℓ^0 is set to

$$A_\ell^0 \triangleq \kappa_0 \cdot R_\ell \cdot |\ell|,$$

where κ_0 is the ITU-R coefficient at frequency f^0 and R_ℓ is the rain-rate equivalent of the original link ℓ (see Section 5.2). For $\mathcal{L}^0$, the attenuation error term J_{atten} is a sum of squares and is therefore convex. We write

$$\hat{R}^0 \leftarrow \text{LBFGSB}(\mathcal{L}^0, P, \hat{R}_{\text{init}})$$

to denote applying L-BFGS-B to the convex surrogate inverse problem. The surrogate objective is convex and includes the strictly convex J_{total}, so it has a single local minimum, which is also the global minimum. Thus, the initial solution $\hat{R}_{\text{init}}$ affects only the convergence speed of L-BFGS-B (In our implementation, $\hat{R}_{\text{init}} = 0.6$ for all pixels.). We refer to the resulting solver as CONVEX SOLVER.

5.5. HOMOTOPY SOLVER: Smooth Transition from Convex to Nonconvex

We extend the virtual-link construction to a parameter $\lambda \in [0, 1]$. For each λ, let

$$\mathcal{L}^\lambda \triangleq \{\ell^\lambda : \ell \in \mathcal{L}\}.$$

The virtual link ℓ^λ keeps the geometry, length, and polarization of ℓ, but has frequency

$$f_\ell^\lambda \triangleq (1-\lambda) f^0 + \lambda f_\ell.$$

Let $(\kappa_\ell(\lambda), \alpha_\ell(\lambda))$ be the ITU-R coefficients at frequency f_ℓ^λ. The attenuation of ℓ^λ is set to

$$A_\ell^\lambda \triangleq |\ell| \cdot \kappa_\ell(\lambda) \cdot R_\ell^{\alpha_\ell(\lambda)}.$$

Algorithm 3: HOMOTOPY SOLVER- a solver that applies a smooth transition from convex optimization to optimizing the nonconvex objective. Assume the same polarization for all the links.

Data: link set $\mathcal{L}$, patch P, initial field $\hat{R}_{\text{init}}$, number of homotopy steps n
Result: rainfall field $\hat{R} : P \to \mathbb{R}_{\geq 0}$

$\hat{R}^0 \leftarrow \text{LBFGSB}(\mathcal{L}^0, P, \hat{R}_{\text{init}})$
for $i = 1, \ldots, n$ **do**
  $\hat{R}^{i/n} \leftarrow \text{LBFGSB}(\mathcal{L}^{i/n}, P, \hat{R}^{(i-1)/n})$
return $\hat{R}^1$

[1] Note that f^0 depends on the link's polarization. For simplicity, we consider a single polarization.

6. Experimental Results and Analysis

We evaluated IDW, ILDW, Solver(ILDW), Convex Solver, Homotopy Solver, and Solver(GT) on all 200 benchmark patches: 100 winter and 100 summer rain events. The ground-truth in these experiments is the rain-rate fields of the benchmark (from which link attenuations were derived).

6.1. Notation

For a patch P and solver s, let $R_P(p)$ denote the ground-truth for pixel p. Let $\widehat{R}_{P,s}(p)$ denote the estimate returned by solver s for the pixel $p \in P$. Define the means:

$$\overline{\widehat{R}}_{P,s} \triangleq \frac{1}{|P|} \sum_{p \in P} \widehat{R}_{P,s}(p), \qquad \overline{R}_P \triangleq \frac{1}{|P|} \sum_{p \in P} R_P(p).$$

For a patch P and solver s, we compute three values: the root mean squared error

$$\mathrm{RMSE}_{P,s} \triangleq \sqrt{\frac{1}{|P|} \sum_{p \in P} \left(\widehat{R}_{P,s}(p) - R_P(p) \right)^2},$$

the signed bias

$$\mathrm{Bias}_{P,s} \triangleq \frac{1}{|P|} \sum_{p \in P} \left(\widehat{R}_{P,s}(p) - R_P(p) \right),$$

and the Pearson correlation coefficient

$$r_{P,s} \triangleq \frac{\sum_{p \in P} \left(\widehat{R}_{P,s}(p) - \overline{\widehat{R}}_{P,s} \right) \left(R_P(p) - \overline{R}_P \right)}{\sqrt{\sum_{p \in P} \left(\widehat{R}_{P,s}(p) - \overline{\widehat{R}}_{P,s} \right)^2} \sqrt{\sum_{p \in P} \left(R_P(p) - \overline{R}_P \right)^2}}.$$

For a pixel p, the absolute error (AE) is defined by

$$AE_{P,s}(p) \triangleq |\widehat{R}_{P,s}(p) - R_P(p)|,$$

and the relative absolute error (RAE) is defined by

$$RAE_{P,s}(p) \triangleq \frac{AE_{P,s}(p)}{R_P(p)}.$$

To avoid division by zero, we use the absolute error for nonrainy pixels.

6.2. RMSE, Bias, and Pearson Correlation Results and Analysis

The overall patchwise reconstruction metrics across all 200 patches are summarized in Table 6. The corresponding metrics stratified by distance to the third-closest link (d_3) are reported in Table 7. The SOLVER(GT) results provide a best-case reference for the nonconvex formulation when the solver is initialized with the ground truth.

Paired t analysis for RMSE

Table 8 summarizes the paired t tests for patchwise RMSE differences relative to IDW. Negative differences indicate that method A has lower RMSE than IDW, which is the one-sided alternative tested here. The paired results show lower RMSE than IDW for all alternatives; ILDW wins in 195 of 200 patches, while CONVEX SOLVER and HOMOTOPY SOLVER win in all 200. Their mean patchwise RMSE reduction relative to IDW is 0.64 mm h^{-1}. Table 8 reports the full paired comparisons.

Method	n	RMSE	NRMSE	Bias	Normalized bias	Pearson r
IDW	200	2.18 ± 1.56	0.590 ± 0.341	0.05 ± 0.32	0.012 ± 0.082	0.835 ± 0.071
ILDW	200	2.07 ± 1.50	0.562 ± 0.328	0.07 ± 0.28	0.017 ± 0.073	0.849 ± 0.065
SOLVER(ILDW)	200	1.94 ± 1.54	0.523 ± 0.348	0.06 ± 0.27	0.016 ± 0.068	0.875 ± 0.074
CONVEX SOLVER	200	1.53 ± 1.17	0.415 ± 0.261	0.05 ± 0.13	0.011 ± 0.031	0.918 ± 0.045
HOMOTOPY SOLVER	200	1.53 ± 1.17	0.415 ± 0.261	0.04 ± 0.13	0.011 ± 0.031	0.918 ± 0.045
SOLVER(GT)	200	0.61 ± 0.49	0.167 ± 0.107	0.01 ± 0.03	0.001 ± 0.007	0.973 ± 0.031

Table 6
Patchwise reconstruction metrics across the full 200-patch benchmark (100 winter and 100 summer patches). Each entry is the mean ± sample SD of one value per patch. For each patch, NRMSE and normalized bias are the RMSE and signed bias divided by the mean ground-truth rain rate over that patch.

Paired t analysis for absolute bias

Table 9 summarizes the paired mean tests for patchwise absolute-bias differences relative to IDW. Negative differences indicate that method A is closer to zero bias than IDW, which is the one-sided alternative tested here. The paired results show that all alternatives significantly reduce absolute bias relative to IDW. The reduction is heterogeneous across patches: CONVEX SOLVER and HOMOTOPY SOLVER have smaller absolute bias than IDW in 157 of 200 patches, reducing mean absolute bias from 0.21 to about 0.09 mm h^{-1}. Table 9 reports the paired comparisons.

Paired comparison of Pearson correlation

Table 10 summarizes patchwise wins for Pearson correlation relative to IDW, where higher r indicates better spatial agreement with the ground-truth rainfall field. The results show that all alternatives improve this measure in most patches. The evidence is strongest for CONVEX SOLVER, HOMOTOPY SOLVER, and SOLVER(GT), which have higher r than IDW in all 200 patches. This indicates that the optimization-based methods improve not only RMSE and bias, but also the spatial agreement of the reconstructed rainfall fields.

6.3. Taylor Diagrams

The per-solver Taylor diagrams show one point per patch, with summer patches in red and winter patches in blue.

6.4. Confusion Summaries

Table 11 summarizes wet/dry classification at the 0.6 mm h^{-1} threshold across all 200 patches. By the fraction-of-all-pixels columns, CONVEX SOLVER reduces false-wet pixels from 0.082 for IDW to 0.047, while false-dry pixels change from 0.004 to 0.005. Among dry ground-truth pixels, the false-wet conditional rate falls from 0.488 to 0.307.

6.5. Attenuation Error

Table 12 summarizes the error between the ground-truth attenuation and the attenuation reconstructed from the estimated rainfall field using the forward model. The main quantity is attenuation MAE, reported in dB km^{-1}. The table also reports the attenuation cost-function term J_{atten}. For a given patch P and solver s, the attenuation MAE per kilometer equals $\frac{1}{|\mathcal{L}_P|}\sum_{\ell\in\mathcal{L}_P}\frac{\left|\hat{A}_{\ell,s}-A_\ell\right|}{|\ell|}$, where $\hat{A}_{\ell,s}$ is the attenuation implied on link ℓ by the ITU-R forward model according to the rain-rate solution $\hat{R}_{P,s}$, A_ℓ is the ground-truth attenuation, and $|\ell|$ is the link length in kilometers.

Across the 200 patches, CONVEX SOLVER and HOMOTOPY SOLVER fit the link attenuations more closely than the interpolation baselines; HOMOTOPY SOLVER has the lowest mean attenuation MAE (1.4×10^{-3} dB km^{-1}). The same ordering is reflected in J_{atten}, as expected because this term is part of the cost function. GE: discuss the error compared to ITU attenuatio

6.6. Median RAE for wet pixels and median AE for dry pixels

Because pixel-level errors within a patch are spatially dependent, we do not treat individual pixels as independent replicates. Instead, for each patch, stratum, and solver, we summarize the pixel-level errors by a single median value, using median RAE for rainy pixels and median AE for non-rainy pixels. The median is used because within-patch error

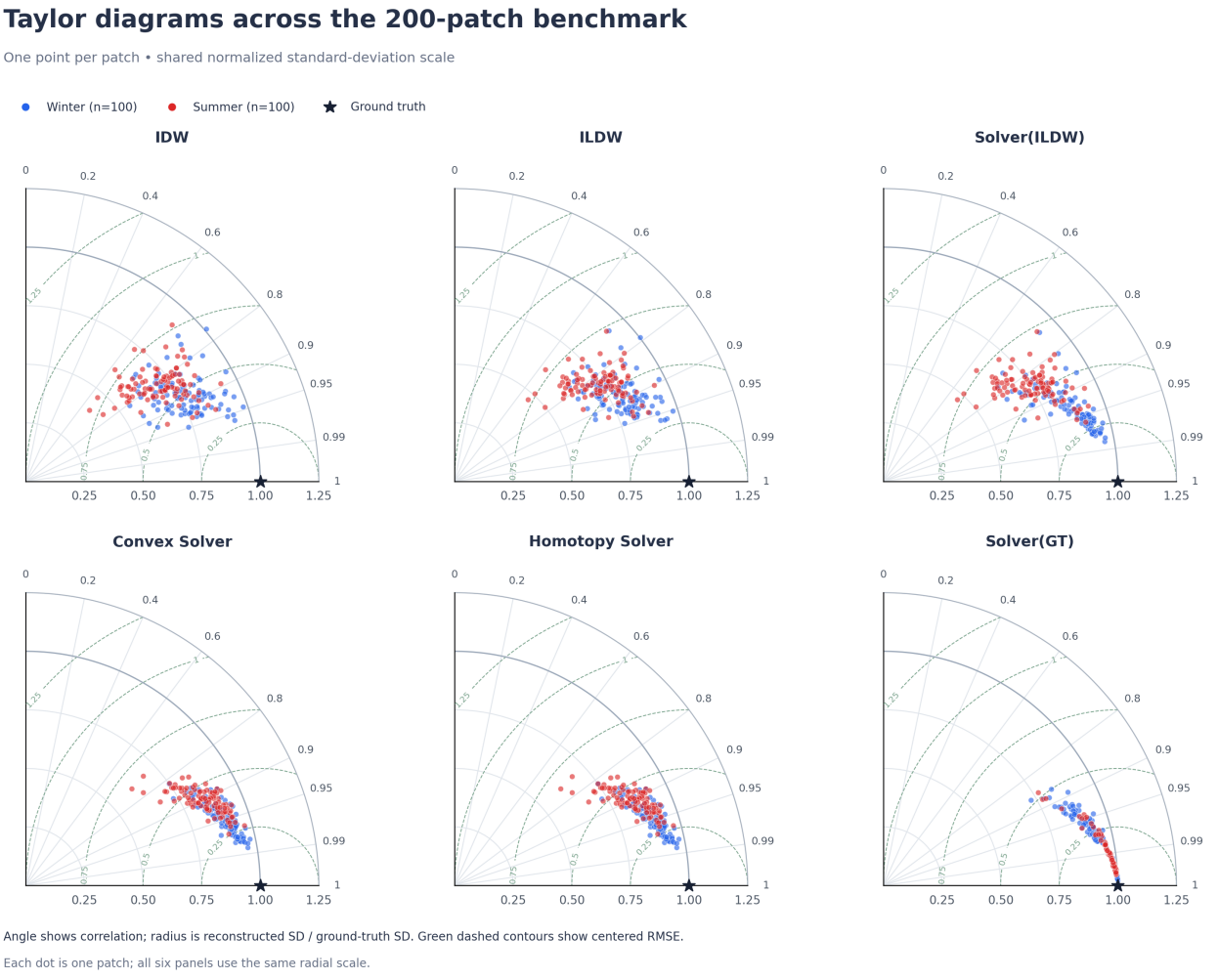


Figure 4: Per-solver Taylor diagrams for the six reconstruction methods, with one point per patch in each panel (winter blue, summer red). The radius is the reconstructed-field standard deviation normalized by the ground-truth standard deviation (ground truth is at radius 1); the angle represents Pearson correlation. Green dashed contours indicate centered RMSE.

distributions are often asymmetric and can have long upper tails or rare large errors, so it provides a robust summary of typical performance within each patch.

We partitioned the pixels in every patch into rainy and non-rainy pixels according to the ground-truth using a threshold of 0.6 mm h^{-1}. For rainy pixels, we compute the relative absolute error, and for non-rainy pixels, we compute the absolute error. We summarize the medians of these results per patch after binning pixels by $d_3(p)$. Figures 5 and 6 depict the resulting medians for various solvers. Table 13 reports the corresponding patchwise win counts for CONVEX SOLVER relative to IDW. CONVEX SOLVER has lower median RAE/AE than IDW in a vast majority of the patches. Some patches have no pixels in strata with large d_3 distances.

7. Case Study Figures and Objective Evolution

In this section, we present two case studies, in which we compare IDW, SOLVER(GT), and CONVEX SOLVER. In the first case study, SOLVER(GT) almost equals the ground-truth while CONVEX SOLVER slightly improves over IDW.

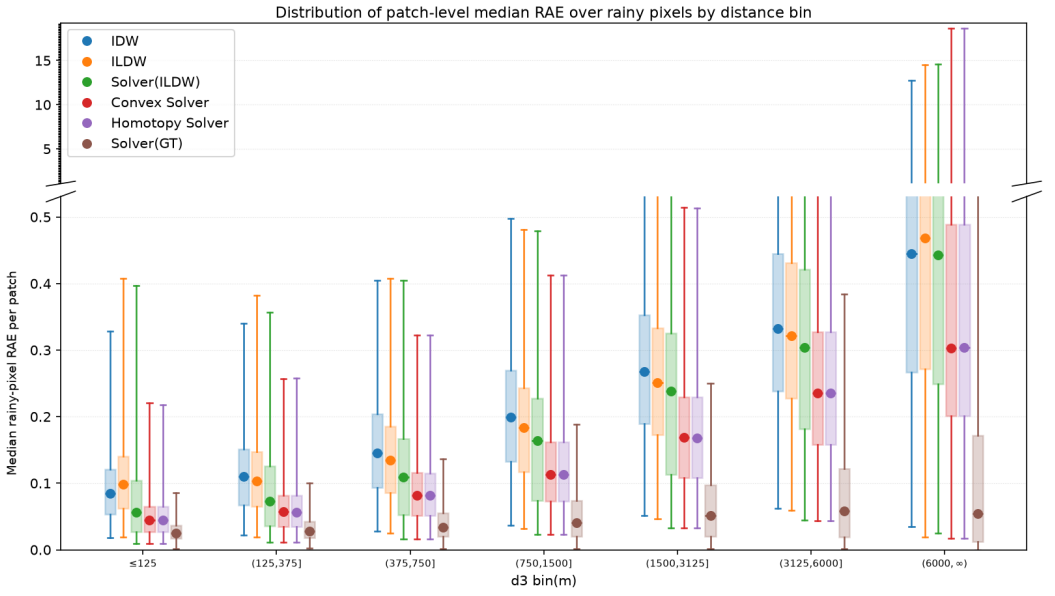


Figure 5: Rainy-pixel d_3-profile box-and-whisker summary across the 200-patch benchmark. Each patch with rainy pixels in a bin contributes one median relative absolute error. Boxes span the 25th to 75th percentiles across patches; whiskers show the minimum and maximum. The number of contributing patches varies by bin, and rainy-pixel counts are summarized in Table 5.

To quantify how often these two qualitative patterns occur, we use the simple ratio

$$q(P) \triangleq \frac{\mathrm{RMSE}_{P,\text{CONVEX SOLVER}}}{\mathrm{RMSE}_{P,\text{SOLVER(GT)}}}.$$

We assign patch P to Case 1 (the Case Study I pattern) when $q(P) \geq 2$ and to Case 2 (the Case Study II pattern) when $q(P) < 2$. Of the 200 benchmark patches, 24 are in Case 1 and 176 are in Case 2. The smallest Case 1 ratio is 2.6310, whereas the largest Case 2 ratio is 1.9767; thus no observed patch lies immediately at the threshold.

For this patch, $J(\text{SOLVER(GT)}) = 1.485$ and $\mathrm{RMSE}(\text{SOLVER(GT)}) = 0.102$, while $J(\text{CONVEX SOLVER}) = 0.265$ and $\mathrm{RMSE}(\text{CONVEX SOLVER}) = 0.702$. This example shows that a lower value of the objective does not necessarily imply a more accurate reconstruction of the ground-truth rainfall field. For Case Study I, Figure 7 shows the reconstructed rainfall fields and relative errors, and Figure 8 shows the evolution of the weighted objective over L-BFGS-B iterations.

In the second case study, SOLVER(GT) and CONVEX SOLVER are visually similar in the reconstruction panels. For this patch, $J(\text{SOLVER(GT)}) = 0.289$ and $\mathrm{RMSE}(\text{SOLVER(GT)}) = 0.552$, while $J(\text{CONVEX SOLVER}) = 0.290$ and $\mathrm{RMSE}(\text{CONVEX SOLVER}) = 0.565$. This example shows a case in which the two solutions have almost the same objective value and almost the same reconstruction error. For Case Study II, Figure 9 shows the reconstructed rainfall fields and relative errors, and Figure 10 shows the evolution of the weighted objective over L-BFGS-B iterations.

8. Discussion

The benchmark isolates the geometric inversion problem and thereby provides a clear setting for method development. At the same time, several sources of uncertainty remain outside the scope of the current construction, including wet-antenna effects, baseline estimation, quantization, mixed-phase precipitation, and temporal dynamics. These factors matter in operational deployments and should be incorporated in later stages once the benchmarked inverse methods are well understood.

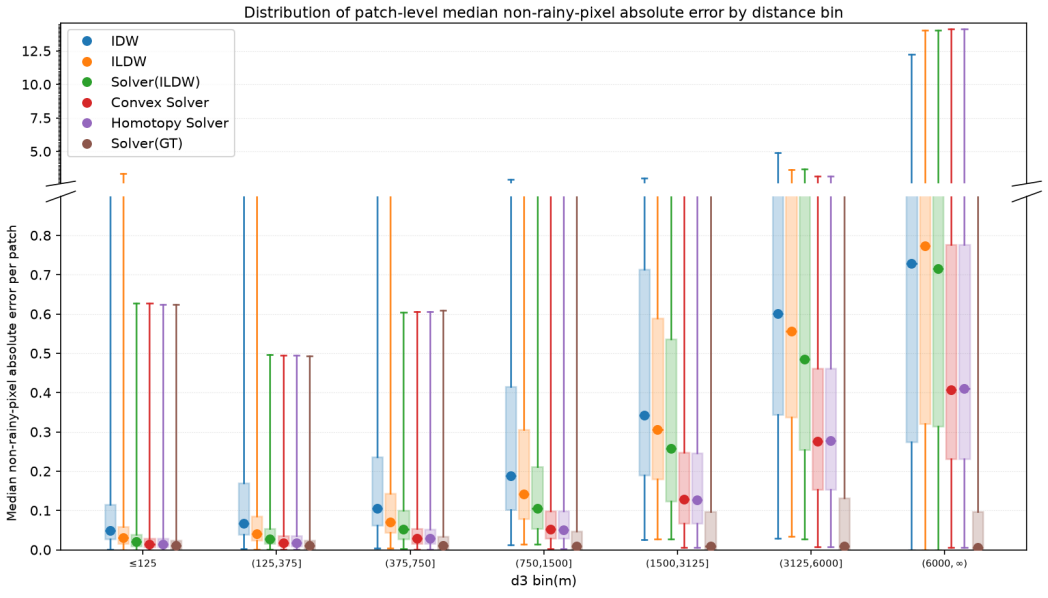


Figure 6: Non-rainy-pixel d_3-profile box-and-whisker summary across the 200-patch benchmark. Each patch with non-rainy pixels in a bin contributes one median absolute error. Boxes span the 25th to 75th percentiles across patches; whiskers show the minimum and maximum. The number of contributing patches varies by bin, and non-rainy-pixel counts are summarized in Table 5.

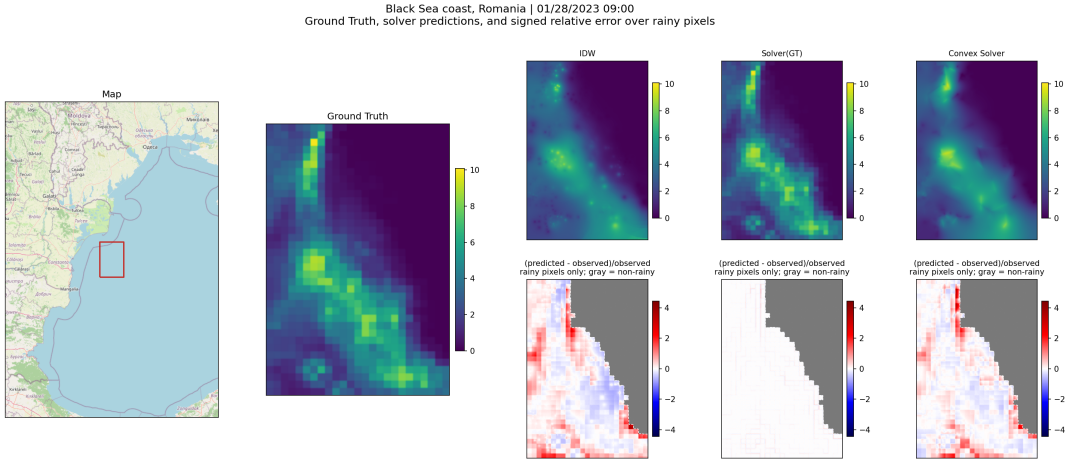


Figure 7: Case Study I: SOLVER(GT) is visually close to the ground truth. The three upper-right panels show the rainfall fields reconstructed by IDW, SOLVER(GT), and CONVEX SOLVER. The corresponding lower-right panels show the relative error of each reconstruction.

The proposed framework also offers interpretability advantages over purely data-driven alternatives. Because the objective function makes the tradeoff between attenuation consistency, spatial smoothness, and rainfall shrinkage explicit, the resulting reconstructions can be inspected and explained in terms of physically meaningful quantities.

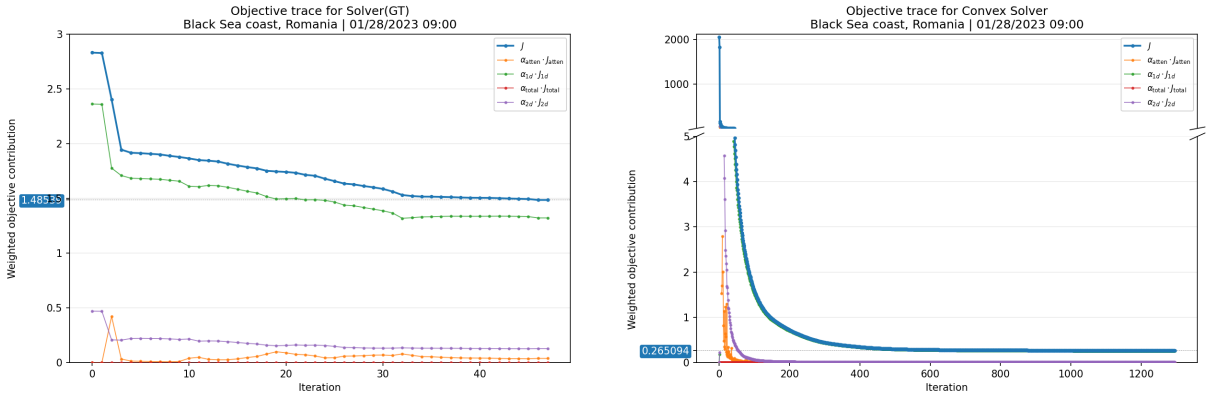


Figure 8: Evolution of the weighted objective over L-BFGS-B iterations for Case Study I. Left: SOLVER(GT), initialized from the ground-truth field, converges to final $J = 1.485$. Right: CONVEX SOLVER reaches a lower final weighted objective, $J = 0.265$, but its reconstruction is farther from the ground truth by RMSE.

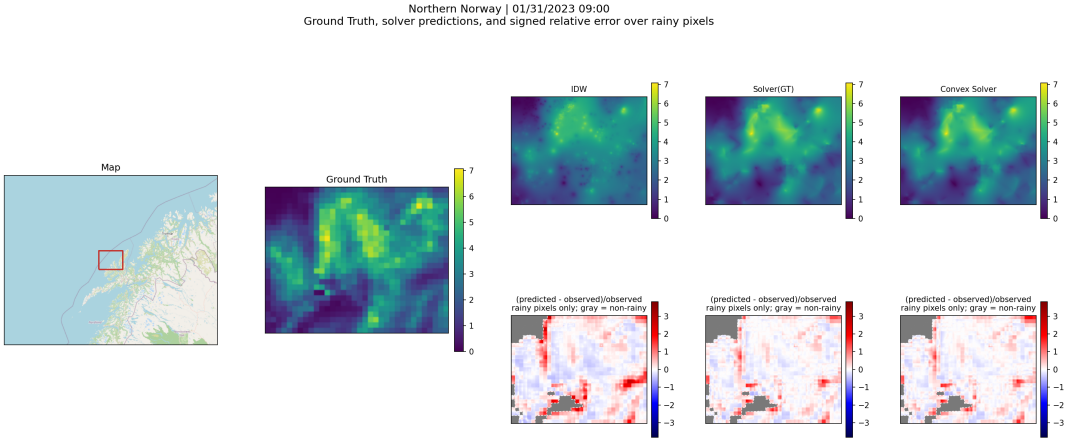


Figure 9: Case Study II: SOLVER(GT) and CONVEX SOLVER produce visually similar reconstructions. The three upper-right panels show the rainfall fields reconstructed by IDW, SOLVER(GT), and CONVEX SOLVER. The corresponding lower-right panels show the relative error of each reconstruction.

The resulting optimization problem is nonconvex due to the nonlinear attenuation model. We therefore compute a locally optimal solution using L-BFGS-B, initialized from the ILDW estimate. In practice, the combination of strong data constraints and regularization yields stable and consistent reconstructions across instances.

The results highlight the importance of the initialization for the nonconvex optimization procedure. In many instances, the ILDW solution lies close to a stationary point of the objective, and the optimizer produces only marginal changes. This indicates that ILDW already achieves a strong balance between attenuation consistency and spatial regularity. Consequently, the role of the nonconvex optimization is primarily to refine the estimate in instances where these objectives are in stronger conflict. In contrast, IDW provides a weaker initialization due to its simplified geometric representation of links, leading to larger discrepancies with the attenuation model.

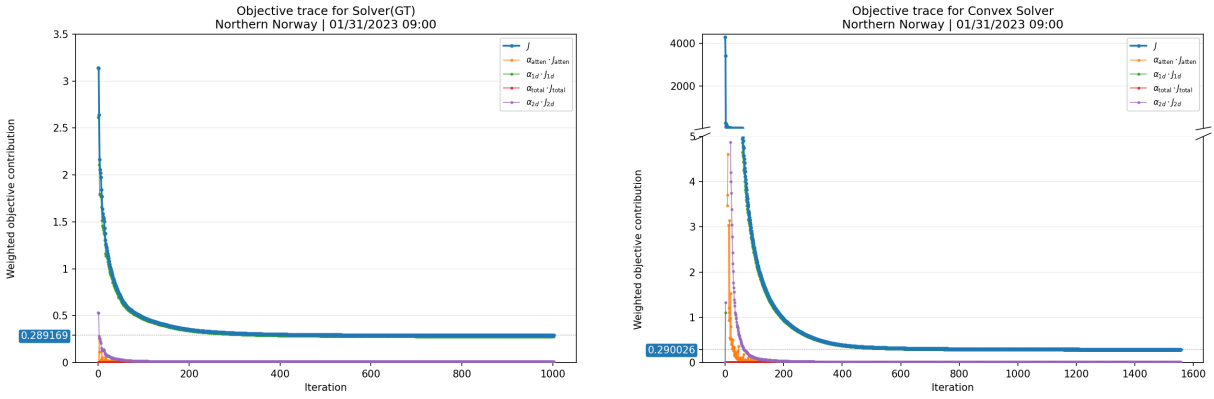


Figure 10: Evolution of the weighted objective over L-BFGS-B iterations for Case Study II. Left: SOLVER(GT), initialized from the ground-truth field, converges to final $J = 0.289$. Right: CONVEX SOLVER reaches a very similar final weighted objective, $J = 0.290$, and a similar RMSE.

9. Conclusion

This journal draft integrates a benchmark-oriented view of rainfall reconstruction from commercial microwave links. Starting from hourly EURADCLIM maps, we extract rain events, simulate CML attenuations through a forward model, and evaluate inverse reconstruction methods against known ground truth. The resulting setup reframes rainfall reconstruction from opportunistic sensing networks as a well-posed inverse problem with an explicit and interpretable objective.

References

Richard H. Byrd, Peihuang Lu, Jorge Nocedal, and Ciyou Zhu. A limited memory algorithm for bound constrained optimization. *SIAM Journal on Scientific Computing*, 16(5):1190–1208, 1995. doi: 10.1137/0916069.

Feng-Wen Chen and Chen-Wuing Liu. Estimation of the spatial rainfall distribution using inverse distance weighting (idw) in the middle of taiwan. *Paddy and Water Environment*, 10(3):209–222, 2012.

Christian Chwala and Harald Kunstmann. Commercial microwave link networks for rainfall observation: assessment of the current status and future challenges. *WIREs Water*, 6(2):e1337, 2019. doi: 10.1002/wat2.1337.

Adam Eshel, Jonatan Ostrometzky, Shani Gat, Pinhas Alpert, and Hagit Messer. Spatial reconstruction of rain fields from wireless telecommunication networks–scenario-dependent analysis of IDW-based algorithms. *IEEE Geoscience and Remote Sensing Letters*, 17(5):770–774, 2020. doi: 10.1109/LGRS.2019.2935348.

Adam Eshel, Hagit Messer, Harald Kunstmann, Pinhas Alpert, and Christian Chwala. Quantitative analysis of the performance of spatial interpolation methods for rainfall estimation using commercial microwave links. *Journal of Hydrometeorology*, 22(4):831–843, 2021.

EUMETNET OPERA. Weather radar information model for implementation with the HDF5 format (ODIM_H5). https://www.eumetnet.eu/wp-content/uploads/2026/01/ODIM_H5_v2.4.2_final.pdf, 2026.

Martin Fencl, Jörg Rieckermann, Petr Sỳkora, David Stránskỳ, and Vojtěch Bareš. Commercial microwave links instead of rain gauges: fiction or reality? *Water Science and Technology*, 71(1):31–37, 2015.

Shani Gat and Hagit Messer. A comparative study of the performance of parameter estimation of a 2-d field using line-and point-projection sensors. In *2019 IEEE 8th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)*, pages 146–150. IEEE, 2019.

Kenrich Lewis Stuart Gunn and Thomas William Russell East. The microwave properties of precipitation particles. *Quart. J. Roy. Meteorol. Soc.*, 80(346):522–545, 1954.

Jin Guo-Dong, Liu Yan-cong, and Niu Wen-Jie. Comparison between inverse distance weighting method and kriging. *Journal of Changchun University of Technology*, 3, 2003.

Congzheng Han, Gaoyuan Zhang, Baofeng Ji, Juan Huo, and Hagit Messer. On the potential of using emerging microwave links for city rainfall monitoring. *IEEE Communications Magazine*, 61(11):174–180, 2023.

ITU-R. Specific attenuation model for rain for use in prediction methods. Recommendation P.838-3, International Telecommunication Union, 2005.

Roy Janco, Jonatan Ostrometzky, and Hagit Messer. Rain estimation from smart citys e-band links. In *2022 IEEE 14th Image, Video, and Multidimensional Signal Processing Workshop (IVMSP)*, pages 1–5. IEEE, 2022.

KNMI. EURADCLIM: the european climatological high-resolution gauge-adjusted radar rainfall dataset. https://www.knmi.nl/research/observations-data-technology/projects/euradclim-the-european-climatological-high-resolution-gauge-adjusted-radar-rai 2023. Project page.

H Leijnse, R Uijlenhoet, and JNM Stricker. Microwave link rainfall estimation: Effects of link length and frequency, temporal sampling, power resolution, and wet antenna attenuation. *Advances in Water Resources*, 31(11):1481–1493, 2008.

Roberto Nebuloni, Filippo Giannetti, Fabiola Sapienza, Vincenzo Lottici, Elisa Adirosi, Giacomo Roversi, Elia Covi, Christian Gianoglio, Matteo Colli, and Carlo De Michele. A review of technical aspects and challenges in opportunistic rainfall estimation using satellite and terrestrial microwave links: How wireless infrastructure can be used for rainfall monitoring. *IEEE Geoscience and Remote Sensing Magazine*, 2025.

R. L. Olsen, D. V. Rogers, and D. B. Hodge. The aRb relation in the calculation of rain attenuation. *IEEE Transactions on Antennas and Propagation*, 26(2):318–329, 1978. doi: 10.1109/TAP.1978.1141845.

Jonatan Ostrometzky and Hagit Messer. Opportunistic weather sensing by smart city wireless communication networks. *Sensors*, 24(24):7901, 2024.

A. Overeem, H. Leijnse, and R. Uijlenhoet. Country-wide rainfall maps from cellular communication networks. *Proceedings of the National Academy of Sciences*, 110.8:2741–2745, 2013.

Aart Overeem, Hidde Leijnse, and Remko Uijlenhoet. Four-year commercial microwave link dataset for the netherlands, 2016.

Aart Overeem, Hidde Leijnse, and Remko Uijlenhoet. EURADCLIM: the european climatological high-resolution gauge-adjusted radar rainfall dataset and its potential for understanding extreme precipitation. *Earth System Science Data*, 15:1441–1460, 2023. doi: 10.5194/essd-15-1441-2023.

Pauli Virtanen et al. Scipy 1.0: Fundamental algorithms for scientific computing in python. *Nature Methods*, 17:261–272, 2020.

Ciyou Zhu, Richard H. Byrd, Jorge Nocedal, and José Luis Morales. Algorithm 778: L-BFGS-B: Fortran subroutines for large-scale bound-constrained optimization. *ACM Transactions on Mathematical Software*, 23(4):550–560, 1997. doi: 10.1145/279232.279236.

Dale Zimmerman, Claire Pavlik, Amy Ruggles, and Marc P Armstrong. An experimental comparison of ordinary and universal kriging and inverse distance weighting. *Mathematical Geology*, 31(4):375–390, 1999.

A. Illustrative Examples

This appendix presents a minimal example that isolates an important property of the inverse problem. Namely, we give a concrete counterexample showing that J_{atten} is nonconvex.

A.1. A 3×3 Counterexample to Convexity

Consider one horizontal link ℓ crossing the three middle-row pixels of a 3×3 grid. The crossed length is 0.125 km in each pixel, so $|\ell| = 0.375$ km. Let the link use horizontal polarization at 38.003 GHz, for which the ITU-R coefficients used here are $\kappa = 0.400171594750$ and $\alpha = 0.881535333268$. For this one-link instance,

$$\widehat{A}(R) = \sum_{p \in \ell} d_p \kappa R(p)^\alpha, \qquad J_{\text{atten}}(R) = \frac{\left(\widehat{A}(R) - A_{\text{obs}}\right)^2}{|\ell|}.$$

Take the two strictly positive fields

$$R^{(L)} = \begin{bmatrix} 1 & 1 & 1 \\ 20 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \qquad R^{(R)} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 20 \\ 1 & 1 & 1 \end{bmatrix} \quad \text{mm h}^{-1}.$$

They are left-right reflections across the grid's vertical centerline. Either field could be the ground truth: the link crosses the reflected pixels over equal lengths, so both fields produce

$$\widehat{A}\left(R^{(L)}\right) = \widehat{A}\left(R^{(R)}\right) = A_{\text{obs}} = 0.801595407171 \text{ dB},$$

and hence $J_{\text{atten}}(R^{(L)}) = J_{\text{atten}}(R^{(R)}) = 0$. Their arithmetic mean is

$$R^{(M)} = \frac{R^{(L)} + R^{(R)}}{2} = \begin{bmatrix} 1 & 1 & 1 \\ 10.5 & 1 & 10.5 \\ 1 & 1 & 1 \end{bmatrix}.$$

For $g(r) = r^\alpha$, we have $g''(r) = \alpha(\alpha-1)r^{\alpha-2} < 0$ because $0 < \alpha < 1$; hence g is strictly concave and $2(10.5)^\alpha > 20^\alpha + 1$. Consequently,

$$\widehat{A}(R^{(M)}) = 0.845082934781 \text{ dB} > A_{\text{obs}}, \qquad J_{\text{atten}}(R^{(M)}) = 0.005043106820 > 0.$$

Therefore,

$$J_{\text{atten}}\left(\frac{R^{(L)} + R^{(R)}}{2}\right) > \frac{J_{\text{atten}}(R^{(L)}) + J_{\text{atten}}(R^{(R)})}{2} = 0,$$

which violates the defining inequality for a convex function.

d_3 bin (m)	Method	n_P (mean pixels)	RMSE	NRMSE	Bias	Normalized bias	Pearson r
≤125	IDW	200 (4958.6)	1.11 ± 0.92	0.301 ± 0.213	-0.04 ± 0.10	-0.010 ± 0.024	0.958 ± 0.023
≤125	ILDW	200 (4958.6)	1.28 ± 1.09	0.346 ± 0.255	0.01 ± 0.07	0.003 ± 0.017	0.939 ± 0.033
≤125	SOLVER(ILDW)	200 (4958.6)	0.92 ± 0.91	0.247 ± 0.221	0.00 ± 0.04	0.001 ± 0.010	0.970 ± 0.027
≤125	CONVEX SOLVER	200 (4958.6)	0.63 ± 0.54	0.171 ± 0.128	0.00 ± 0.02	0.000 ± 0.005	0.985 ± 0.011
≤125	HOMOTOPY SOLVER	200 (4958.6)	0.63 ± 0.53	0.171 ± 0.128	0.00 ± 0.02	0.000 ± 0.005	0.985 ± 0.011
≤125	SOLVER(GT)	200 (4958.6)	0.39 ± 0.34	0.106 ± 0.073	0.00 ± 0.01	0.000 ± 0.002	0.992 ± 0.007
(125,375]	IDW	200 (19536.2)	1.25 ± 1.01	0.338 ± 0.233	-0.04 ± 0.10	-0.010 ± 0.025	0.948 ± 0.026
(125,375]	ILDW	200 (19536.2)	1.23 ± 1.02	0.333 ± 0.234	-0.01 ± 0.05	-0.002 ± 0.012	0.946 ± 0.028
(125,375]	SOLVER(ILDW)	200 (19536.2)	1.02 ± 0.97	0.275 ± 0.231	-0.01 ± 0.04	-0.002 ± 0.009	0.965 ± 0.028
(125,375]	CONVEX SOLVER	200 (19536.2)	0.73 ± 0.61	0.198 ± 0.142	0.00 ± 0.02	0.000 ± 0.005	0.981 ± 0.012
(125,375]	HOMOTOPY SOLVER	200 (19536.2)	0.73 ± 0.61	0.198 ± 0.142	0.00 ± 0.02	0.000 ± 0.005	0.981 ± 0.012
(125,375]	SOLVER(GT)	200 (19536.2)	0.43 ± 0.38	0.117 ± 0.079	0.00 ± 0.01	0.000 ± 0.003	0.991 ± 0.008
(375,750]	IDW	200 (31246.2)	1.53 ± 1.20	0.416 ± 0.278	-0.03 ± 0.17	-0.009 ± 0.042	0.926 ± 0.037
(375,750]	ILDW	200 (31246.2)	1.40 ± 1.12	0.379 ± 0.258	-0.01 ± 0.12	-0.004 ± 0.028	0.935 ± 0.032
(375,750]	SOLVER(ILDW)	200 (31246.2)	1.25 ± 1.12	0.340 ± 0.266	-0.01 ± 0.10	-0.003 ± 0.024	0.950 ± 0.035
(375,750]	CONVEX SOLVER	200 (31246.2)	0.95 ± 0.78	0.258 ± 0.180	0.00 ± 0.04	0.001 ± 0.009	0.969 ± 0.019
(375,750]	HOMOTOPY SOLVER	200 (31246.2)	0.95 ± 0.78	0.258 ± 0.180	0.00 ± 0.04	0.001 ± 0.009	0.969 ± 0.019
(375,750]	SOLVER(GT)	200 (31246.2)	0.50 ± 0.43	0.136 ± 0.091	0.00 ± 0.01	0.000 ± 0.004	0.987 ± 0.013
(750,1500]	IDW	200 (60771.3)	1.88 ± 1.41	0.512 ± 0.316	-0.04 ± 0.25	-0.011 ± 0.061	0.885 ± 0.056
(750,1500]	ILDW	200 (60771.3)	1.74 ± 1.31	0.472 ± 0.294	-0.02 ± 0.19	-0.007 ± 0.049	0.900 ± 0.048
(750,1500]	SOLVER(ILDW)	200 (60771.3)	1.60 ± 1.34	0.433 ± 0.311	-0.02 ± 0.18	-0.005 ± 0.044	0.919 ± 0.054
(750,1500]	CONVEX SOLVER	200 (60771.3)	1.24 ± 0.98	0.335 ± 0.219	0.00 ± 0.06	0.000 ± 0.015	0.949 ± 0.029
(750,1500]	HOMOTOPY SOLVER	200 (60771.3)	1.24 ± 0.98	0.335 ± 0.219	0.00 ± 0.06	0.000 ± 0.015	0.949 ± 0.029
(750,1500]	SOLVER(GT)	200 (60771.3)	0.57 ± 0.49	0.155 ± 0.107	0.00 ± 0.02	-0.001 ± 0.007	0.980 ± 0.022
(1500,3125]	IDW	200 (104120.1)	2.37 ± 1.72	0.641 ± 0.375	0.00 ± 0.40	0.000 ± 0.105	0.807 ± 0.089
(1500,3125]	ILDW	200 (104120.1)	2.26 ± 1.65	0.610 ± 0.361	0.02 ± 0.36	0.004 ± 0.095	0.826 ± 0.081
(1500,3125]	SOLVER(ILDW)	200 (104120.1)	2.12 ± 1.71	0.570 ± 0.386	0.02 ± 0.34	0.005 ± 0.089	0.854 ± 0.093
(1500,3125]	CONVEX SOLVER	200 (104120.1)	1.66 ± 1.29	0.450 ± 0.287	0.02 ± 0.15	0.005 ± 0.038	0.907 ± 0.051
(1500,3125]	HOMOTOPY SOLVER	200 (104120.1)	1.66 ± 1.29	0.450 ± 0.287	0.02 ± 0.15	0.005 ± 0.038	0.907 ± 0.051
(1500,3125]	SOLVER(GT)	200 (104120.1)	0.65 ± 0.54	0.178 ± 0.120	0.00 ± 0.04	0.000 ± 0.011	0.969 ± 0.036
(3125,6000]	IDW	200 (54469.9)	2.62 ± 1.91	0.709 ± 0.417	0.25 ± 0.69	0.063 ± 0.172	0.716 ± 0.147
(3125,6000]	ILDW	200 (54469.9)	2.54 ± 1.86	0.688 ± 0.407	0.25 ± 0.65	0.064 ± 0.163	0.734 ± 0.136
(3125,6000]	SOLVER(ILDW)	200 (54469.9)	2.42 ± 1.92	0.652 ± 0.430	0.23 ± 0.63	0.059 ± 0.157	0.769 ± 0.148
(3125,6000]	CONVEX SOLVER	200 (54469.9)	1.97 ± 1.55	0.535 ± 0.349	0.14 ± 0.36	0.036 ± 0.088	0.841 ± 0.099
(3125,6000]	HOMOTOPY SOLVER	200 (54469.9)	1.97 ± 1.55	0.535 ± 0.349	0.14 ± 0.36	0.036 ± 0.088	0.841 ± 0.099
(3125,6000]	SOLVER(GT)	200 (54469.9)	0.66 ± 0.55	0.183 ± 0.127	0.02 ± 0.06	0.006 ± 0.019	0.956 ± 0.057
(6000,9000]	IDW	200 (4946.0)	2.20 ± 1.81	0.596 ± 0.382	0.89 ± 1.22	0.233 ± 0.283	0.582 ± 0.282
(6000,9000]	ILDW	200 (4946.0)	2.17 ± 1.77	0.588 ± 0.370	0.90 ± 1.22	0.236 ± 0.279	0.590 ± 0.280
(6000,9000]	SOLVER(ILDW)	200 (4946.0)	2.09 ± 1.80	0.564 ± 0.379	0.86 ± 1.21	0.223 ± 0.275	0.623 ± 0.277
(6000,9000]	CONVEX SOLVER	200 (4946.0)	1.62 ± 1.58	0.445 ± 0.353	0.43 ± 0.98	0.112 ± 0.234	0.716 ± 0.235
(6000,9000]	HOMOTOPY SOLVER	200 (4946.0)	1.63 ± 1.58	0.445 ± 0.353	0.44 ± 0.98	0.113 ± 0.234	0.716 ± 0.235
(6000,9000]	SOLVER(GT)	200 (4946.0)	0.47 ± 0.44	0.135 ± 0.115	0.10 ± 0.21	0.031 ± 0.061	0.922 ± 0.143
>9000	IDW	119 (600.7)	1.75 ± 1.93	0.469 ± 0.396	1.19 ± 2.05	0.290 ± 0.456	0.320 ± 0.547
>9000	ILDW	119 (600.7)	1.77 ± 2.03	0.469 ± 0.397	1.26 ± 2.15	0.305 ± 0.453	0.292 ± 0.567
>9000	SOLVER(ILDW)	119 (600.7)	1.74 ± 2.06	0.457 ± 0.404	1.25 ± 2.17	0.301 ± 0.455	0.284 ± 0.568
>9000	CONVEX SOLVER	119 (600.7)	1.13 ± 1.77	0.314 ± 0.400	0.58 ± 1.87	0.138 ± 0.438	0.378 ± 0.590
>9000	HOMOTOPY SOLVER	119 (600.7)	1.13 ± 1.77	0.314 ± 0.400	0.58 ± 1.87	0.139 ± 0.438	0.378 ± 0.592
>9000	SOLVER(GT)	119 (600.7)	0.25 ± 0.30	0.076 ± 0.095	0.13 ± 0.27	0.039 ± 0.084	0.802 ± 0.356

Table 7

Patchwise reconstruction metrics stratified by distance to the third-closest link (d_3), across the 200-patch benchmark. Each metric is computed over all pixels in a patch's distance bin; entries are the mean ± sample SD across patches with pixels in that bin. NRMSE and normalized bias use each contributing patch's mean ground-truth rain rate over the full patch as the denominator. n_P is the number of contributing patches; mean pixels is averaged over all 200 patches, including empty bins.

Comparison	n	wins A:IDW	mean A	mean IDW	mean diff	SD diff	SE	t	one-sided p	99% upper bound	decision
ILDW- IDW	200	195:5	2.07	2.18	-0.10	0.09	0.01	-15.83	2.33×10^{-37}	-0.09	reject
SOLVER(ILDW)- IDW	200	196:4	1.94	2.18	-0.24	0.20	0.01	-17.16	2.11×10^{-41}	-0.20	reject
CONVEX SOLVER- IDW	200	200:0	1.53	2.18	-0.64	0.48	0.03	-19.13	2.93×10^{-47}	-0.56	reject
HOMOTOPY SOLVER- IDW	200	200:0	1.53	2.18	-0.64	0.47	0.03	-19.12	2.96×10^{-47}	-0.56	reject
SOLVER(GT)- IDW	200	200:0	0.61	2.18	-1.56	1.37	0.10	-16.13	2.79×10^{-38}	-1.34	reject

Table 8
One-sided paired t tests of patchwise RMSE differences against IDW across all 200 patches. Negative differences favor method A; upper bounds are one-sided 99% bounds.

Comparison	n	wins A:IDW	mean A	mean IDW	mean diff	SD diff	SE	t	one-sided p	99% upper bound	decision
ILDW- IDW	200	138:62	0.19	0.21	-0.02	0.05	0.00	-6.59	1.91×10^{-10}	-0.02	reject
SOLVER(ILDW)- IDW	200	149:51	0.17	0.21	-0.05	0.09	0.01	-7.29	3.63×10^{-12}	-0.03	reject
CONVEX SOLVER- IDW	200	157:43	0.09	0.21	-0.12	0.20	0.01	-8.92	1.50×10^{-16}	-0.09	reject
HOMOTOPY SOLVER- IDW	200	157:43	0.09	0.21	-0.13	0.20	0.01	-8.94	1.36×10^{-16}	-0.09	reject
SOLVER(GT)- IDW	200	183:17	0.01	0.21	-0.20	0.25	0.02	-11.49	4.68×10^{-24}	-0.16	reject

Table 9
One-sided paired t tests of patchwise absolute-bias differences against IDW across all 200 patches. Negative differences favor method A.

Comparison	wins A:IDW	n
ILDW- IDW	190:10	200
SOLVER(ILDW)- IDW	194:6	200
CONVEX SOLVER- IDW	200:0	200
HOMOTOPY SOLVER- IDW	200:0	200
SOLVER(GT)- IDW	200:0	200

Table 10
Patchwise Pearson correlation wins over IDW across all 200 patches.

Method	n	Fraction of all pixels				Conditional rate			
		True wet	False wet	True dry	False dry	True wet	False wet	True dry	False dry
IDW	200	0.799	0.082	0.116	0.004	0.994	0.488	0.442	0.006
ILDW	200	0.798	0.076	0.122	0.004	0.994	0.456	0.474	0.006
SOLVER(ILDW)	200	0.798	0.072	0.125	0.004	0.994	0.404	0.526	0.006
CONVEX SOLVER	200	0.797	0.047	0.150	0.005	0.992	0.307	0.623	0.008
HOMOTOPY SOLVER	200	0.797	0.047	0.150	0.005	0.992	0.308	0.622	0.008
SOLVER(GT)	200	0.800	0.013	0.185	0.003	0.996	0.176	0.754	0.004

Table 11
Patch-averaged wet/dry confusion at the 0.6 mm/h threshold across all 200 patches. Fraction columns are normalized by each patch's pixel count; conditional TP/FN rates use wet GT pixels and FP/TN rates use dry GT pixels.

Solver	n	Attenuation MAE ($\mathrm{dB\,km^{-1}}$)	J_{atten}
IDW	200	0.0819 ± 0.0551	0.1068 ± 0.1508
ILDW	200	0.0222 ± 0.0158	0.0089 ± 0.0169
SOLVER(ILDW)	200	0.0045 ± 0.0064	0.0007 ± 0.0021
CONVEX SOLVER	200	0.0021 ± 0.0021	0.0001 ± 0.0002
HOMOTOPY SOLVER	200	0.0014 ± 0.0014	0.0000 ± 0.0001

Table 12
Patch-averaged attenuation errors across the 200-patch benchmark. Methods are included when a matching winter and summer solver output is available.

d_3 bin (m)	Rainy pixels, Convex:IDW	Non-rainy pixels, Convex:IDW	n
≤125	200:0	175:1	176
(125,375]	200:0	180:1	181
(375,750]	200:0	182:0	182
(750,1500]	200:0	183:0	183
(1500,3125]	200:0	186:0	186
(3125,6000]	199:1	174:10	184
(6000,9000]	151:49	116:47	163
>9000	72:19	47:28	75

Table 13
Patchwise median-error wins by d_3 bin across all 200 patches. Rainy pixels are compared by median relative absolute error; non-rainy pixels by median absolute error. n is the number of paired patches with pixels in both strata.